

Miltiadis Kofinas

DEEP LEARNING RESEARCH SCIENTIST

Amsterdam, The Netherlands

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Education

Postdoc in Artificial Intelligence for Climate

VRIJE UNIVERSITEIT AMSTERDAM (VU)

- Title: Foundation Models for Weather Forecasting
- Advisor: Dim Coumou

📍 Amsterdam, The Netherlands

Feb. 2025 - now

PhD in Computer Science (Artificial Intelligence)

UNIVERSITY OF AMSTERDAM (UvA)

- Title: Deep Future Spatio-temporal Forecasting
- Supervisor: Efstratios Gavves
- Expected graduation: 2026

📍 Amsterdam, The Netherlands

Apr. 2020 - June 2024

Diploma (M.Sc. equivalent) in Electrical and Computer Engineering

ARISTOTLE UNIVERSITY OF THESSALONIKI (AUTH)

- Specialization Field: Electronics and Computer Engineering
- GPA: 7.57/10
- ECTS: 307

📍 Thessaloniki, Greece

Oct. 2010 - Nov. 2018

- Thesis: Scene Graph Generation using Message Passing Neural Networks and Graph Convolutional Networks
 - SUPERVISORS: POSTDOCTORAL RESEARCH ASSOCIATE CHRISTOS DIOU & ASSOCIATE PROFESSOR ANASTASIOS DELOPOULOS
 - Visual scene graph generation using an end-to-end neural network that incorporates a message passing neural network, propagating contextual information between objects and their relationships to iteratively refine its predictions, as well as a relationship pruning network that learns to identify and dismiss unlikely relationships.
 - Links to thesis: [Greek \(Original\)](#), [English \(Translated\)](#)

Research Experience

Research Assistant

UNIVERSITY OF AMSTERDAM (UvA)

Project: Scene Graph Generation using Graph Transformer Networks

Supervisors: Assistant Professor Efstratios Gavves & Professor Cees G.M. Snoek

- Development of a novel Graph Network for visual scene graph generation that explicitly utilizes both local and global information on the graph space, using Transformer blocks to attend to global context.

📍 Amsterdam, The Netherlands

Mar. 2019 - May 2019

KEYWORDS: [VISUAL SCENE GRAPH GENERATION](#) · [GRAPH NEURAL NETWORKS](#) · [TRANSFORMERS](#) · [GRAPH PRUNING](#)

Computer Vision & Machine Learning Engineer

P.A.N.D.O.R.A. ROBOTICS TEAM, ARISTOTLE UNIVERSITY OF THESSALONIKI

- Development of a general-purpose image classification API using RGB-D sensor data, as well as a benchmarking API for performance evaluation of computer vision algorithms; motion detection and obstacle detection from RGB-D sensor data.
- Honors: 2nd Best Autonomous Robot, Robocup Rescue Competition, Hefei, China, July 2015

📍 Thessaloniki, Greece

Oct. 2014 - Oct. 2015

KEYWORDS: [IMAGE CLASSIFICATION](#) · [NEURAL NETWORKS](#) · [SVMS](#) · [BENCHMARKING](#) · [MOTION DETECTION](#) · [OBSTACLE DETECTION](#)

Reviewer

- I have served as reviewer for the following venues: [ICLR 2026](#), [ICML 2025](#), [ICLR 2025](#), [LoG 2023](#), [NeurIPS 2023](#), [CVPR 2023](#), [ICLR 2023](#), [LoG 2022](#), [CDS 2022](#), [CVIU 2022](#), [PAMI 2022](#), [ICML 2021](#), [ECCV 2020](#).

Technical Skills

Programming Languages Python, C++, C, MATLAB/Octave, Java

Deep Learning Frameworks PyTorch, TensorFlow

Deep Learning Libraries PyTorch Geometric, PyTorch Lightning, WandB, Tensorboard, Hydra

Miscellaneous Git, Linux, SLURM, $\LaTeX$, TikZ, OpenCV, ROS

Selected Publications

CONFERENCE PAPERS

- Yongtuo Liu, Sara Magliacane, Miltiadis Kofinas, and Efstratios Gavves. **Amortized Equation Discovery in Hybrid Dynamical Systems**. In: *The Forty-first International Conference on Machine Learning (ICML)*. 2024 ([ArXiv](#)) ([Github](#))
- Samuele Papa, Riccardo Valperga, David M. Knigge, Miltiadis Kofinas, Phillip Lippe, Jan-jakob Sonke, and Efstratios Gavves. **How to Train Neural Field Representations: A Comprehensive Study and Benchmark**. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024 ([ArXiv](#)) ([Github](#))

- [Miltiadis Kofinas](#)[†], Boris Knyazev, Yan Zhang, Yunlu Chen, Gertjan J. Burghouts, Efstratios Gavves, Cees G.M. Snoek, and David W. Zhang[†]. **Graph Neural Networks for Learning Equivariant Representations of Neural Networks**. In: *12th International Conference on Learning Representations (ICLR)*. 2024 ([ArXiv](#)) ([OpenReview](#)) ([Github](#)) [[Oral](#)] [†]: Joint first and last authors
- [Miltiadis Kofinas](#), Erik J. Bekkers, Naveen Shankar Nagaraja, and Efstratios Gavves. **Latent Field Discovery in Interacting Dynamical Systems with Neural Fields**. In: *Advances in Neural Information Processing Systems 36 (NeurIPS)*. 2023 ([ArXiv](#)) ([OpenReview](#)) ([Github](#))
- Yongtuo Liu, Sara Magliacane, [Miltiadis Kofinas](#), and Efstratios Gavves. **Graph Switching Dynamical Systems**. In: *The Fortieth International Conference on Machine Learning (ICML)*. 2023 ([ArXiv](#)) ([Github](#))
- [Miltiadis Kofinas](#), Naveen Shankar Nagaraja, and Efstratios Gavves. **Roto-translated Local Coordinate Frames For Interacting Dynamical Systems**. In: *Advances in Neural Information Processing Systems 34 (NeurIPS)*. 2021 ([ArXiv](#)) ([OpenReview](#)) ([Github](#))

WORKSHOP PAPERS

- Aviv Shamsian[†], David W. Zhang[†], Aviv Navon, Yan Zhang, [Miltiadis Kofinas](#), Idan Achituve, Riccardo Valperga, Gertjan Burghouts, Efstratios Gavves, Cees Snoek, Ethan Fetaya, Gal Chechik, and Haggai Maron. **Data Augmentations in Deep Weight Spaces**. In: *Workshop on Symmetry and Geometry in Neural Representations (NeurReps)*, *NeurIPS*. 2023 ([ArXiv](#)) [[Oral](#)] [†]: Equal contribution
- Piyush Bagad[†], Floor Eijkelboom[†], Mark Fokkema[†], Danilo de Goede[†], Paul Hilders[†], and [Miltiadis Kofinas](#). **C-3PO: Towards Rotation Equivariant Feature Detection and Description**. In: *3rd Visual Inductive Priors for Data-Efficient Deep Learning Workshop*. 2022 ([OpenReview](#)) [[Oral](#)] [†]: Equal contribution

PREPRINTS

- [Miltiadis Kofinas](#), Samuele Papa, and Efstratios Gavves. **From MLP to NeoMLP: Leveraging Self-Attention for Neural Fields**. 2024 ([ArXiv](#)) ([Github](#))

Teaching Experience

TEACHING ASSISTANT

Machine Learning I	University of Amsterdam, MSc AI	2020, 2021
Deep Learning	University of Amsterdam, MSc AI	2020
Deep Learning II	University of Amsterdam, MSc AI	2022, 2023

Supervision

PHD SUPERVISION

Eliot Walt (Co-supervised with Dim Coumou)
DEEP LEARNING FOR EXTENDED-RANGE WEATHER FORECASTING

Vrije Universiteit Amsterdam, Institute for Environmental Studies (IVM)
Feb. 2025 - now

THESIS SUPERVISION

Kenji Belderbos (Co-supervised with Melinda Galfi)
DEEPARCHETYPES: DATA-DRIVEN DISCOVERY OF EXTREME LONG-LASTING ATMOSPHERIC PATTERNS

Vrije Universiteit Amsterdam, MSc AI
February 2026 - now

Mark Mitrani (Co-supervised with Melinda Galfi)
PREDICTING WEATHER-DRIVEN CROP FAILURES WITH SPACE-TIME TRANSFORMERS

Vrije Universiteit Amsterdam, MSc AI
March 2025 - November 2025

Mikołaj Magiera
INTERPRETATION AND VISUALIZATION OF MEDIUM-RANGE WEATHER FORECASTS THROUGH ATTENTION HEATMAPS

Vrije Universiteit Amsterdam, BSc CS
Apr. 2025 - July 2025

Constantinos Dimitriou
REGIONAL MEDIUM-RANGE FORECASTING

Vrije Universiteit Amsterdam, BSc AI
Apr. 2025 - July 2025

Danilo Rijpstra
LATENT SPACE ANALYSIS ON HEATWAVE CLUSTERING/DETECTION

Vrije Universiteit Amsterdam, BSc AI
Apr. 2025 - July 2025

Kevyn M. Smit
EVALUATING FINE-TUNING METHODS FOR AURORA'S S2S FORECASTING: A COMPARISON OF LoRA AND FOURIERFT

Vrije Universiteit Amsterdam, BSc AI
Apr. 2025 - July 2025

Victor Kyriakou
EQUIVARIANT TRAJECTORY FORECASTING WITH LATENT ORIENTATION GRAPH NETWORKS

University of Amsterdam, MSc AI
Nov. 2022 - July 2023

Daniel Perez Jensen
PREDICTING RIVER FLOW IN ATACAMA REGION WATERSHEDS

University of Amsterdam, MSc AI
Nov. 2021 - July 2022

Daniël (Stijn) Hamerslag
DRIVING ON DATA, OBJECT DETECTION IN URBAN DRIVING SCENES

University of Amsterdam, BSc AI
Oct. 2020 - Jan. 2021

Talks

Learning on Graphs Conference Amsterdam Meetup
FROM MLP TO NEOMLP: LEVERAGING SELF-ATTENTION FOR NEURAL FIELDS

 *Elsevier, Amsterdam*
27 November, 2024

Learning on Graphs Conference Amsterdam Meetup
NEURAL NETWORKS ARE GRAPHS! GRAPH NEURAL NETWORKS FOR EQUIVARIANT PROCESSING OF NEURAL NETWORKS

 *Elsevier, Amsterdam*
29 November, 2023

Geometric Deep Learning Study Visit

ROTO-TRANSLATED LOCAL COORDINATE FRAMES FOR INTERACTING DYNAMICAL SYSTEMS

 [Vrije Universiteit, Amsterdam](#)
2 June, 2022

Amsterdam Applied ML Meetup

ROTO-TRANSLATED LOCAL COORDINATE FRAMES FOR INTERACTING DYNAMICAL SYSTEMS

 [Hyperion Lab, Amsterdam](#)
6 Apr, 2022

LoGaG: Learning on Graphs and Geometry Reading Group

ROTO-TRANSLATED LOCAL COORDINATE FRAMES FOR INTERACTING DYNAMICAL SYSTEMS

 [Virtual](#)
1 Feb, 2022

— [Recording \(YouTube\)](#)

Languages

Greek Native Language

English Certificate of Proficiency in English, University of Michigan

[Level C2](#)

French Diplôme d'études en langue française B2, Centre international d'études pédagogiques (CIEP)

[Level B2](#)